

Exploring Primordial Black Holes: Formation Dynamics and Mass Spectrum in an Early Matter-Dominated Era

Introduction

Primordial black holes (PBHs) are objects formed in the early universe from the collapse of large density fluctuations, distinct from stellar black holes. Their formation depends heavily on the conditions at different stages of the universe's evolution. In particular, the mass spectrum of PBH, reflecting how many black holes form at each mass, is influenced by whether they formed during the radiation-dominated or matter-dominated eras. This proposal discusses the dynamics of PBH formation, focusing on how perturbations relative to the horizon radius, pressure dynamics, and the Gaussian distribution of perturbations affect the mass spectrum and subsequent evolution of PBHs. During both the radiation-dominated and matter-dominated eras, perturbations needed to exceed a certain threshold for gravitational collapse to occur. This threshold is often described in terms of the density contrast, $\delta\rho/\rho$, which must reach an $O(0.1)$ fraction of unity for collapse. The size of these perturbations relative to the horizon radius is crucial as well. In the radiation-dominated era, the existence of the strong pressure from radiation meant that only perturbations on scales smaller than the horizon radius could collapse into PBHs. If the perturbation scale is smaller than the horizon, pressure forces dominate, preventing collapse. On the other hand, in the matter-dominated era, where pressure is negligible, even smaller-scale perturbations within the horizon can collapse, though they must still be significant enough to overcome gravitational instability. The growth of PBHs after their formation depends on the surrounding environment and the type of accretion they undergo. Initially, PBHs may grow through Bondi accretion, where they pull in surrounding matter in a spherical formation. In a matter dominated phase, this can increase their mass significantly, by factors of 10, 1000, or more. However, as their mass increases, they may transition to Eddington accretion, where a rotating disk forms around the black hole and infall of matter is not as efficient. The conditions for this transition depend on the density of the surrounding medium and the black hole's mass, with larger PBHs more likely to experience Eddington-limited growth. The focus of our research is to further investigate the mass spectrum of PBHs formed during an epoch of early matter domination within the first second after the big bang.

Schedule

September 2024

- Expand thesis outline
- Ensure clarity on key objectives and scope of our project
- Ensure clear understanding on all of the topics of our project

October 2024

- Develop and refine models for PBH formation in the radiation dominated and matter dominated phases
- Finalise equations for density perturbations, collapse conditions, and accretion dynamics
- Draft the theoretical framework chapter

November 2024

- Focusing on modeling PBH formation during the radiation dominated era
- Run calculations on mass spectrum, pressure dynamics, and threshold for gravitational collapse

- Initial results and draft of PBH formation during radiation dominated era

December 2024

- Develop models for PBH formation during the matter dominated era, including accretion dynamics and gravitational instability
- Compare with radiation dominated results

January 2025

- Refine models based on results
- Computational simulations or further analysis

February 2025 - March 2025

- Investigating implications for the PBH mass spectrum, accretion, and long term evolution for both the matter dominated and radiation dominated eras

April 2025

- Study PBHs evaporating via Hawking radiation and the gamma ray observations that constrain their population
- Explore the significance of PBHs in the $\sim 10^{16}$ grams range and their relevance today

May 2025

- Complete conclusion and summarise key points
- Final proofreading
- Submit final proposal

References

- [1] Carr, B., & Green, A. (2024). The History of Primordial Black Holes.
<https://arxiv.org/pdf/2406.05736>
- [2] De Luca, V., & Franciolini, G. (2020, August). *Constraints on primordial black holes: The importance of accretion*. PHYSICAL REVIEW D 102.
https://iris.uniroma1.it/bitstream/11573/1478878/1/De-Luca_Constraints_2020.pdf
- [3] NASA. (n.d.). *Types - NASA science*. NASA. <https://science.nasa.gov/universe/black-holes/types/>
- [4] Ritchie, K. (2019, September). Decay of coherent scalar field oscillations.
https://physics.unm.edu/pandaweb/undergraduate/programs/honorsThesis/2020_ritchie.pdf